Statistical Inference on Personal Health Data: The Effect of Physical Activity on REM Sleep

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***Abstract*—This project investigates the relationship between physical activity (daily step count) and REM sleep duration using personal health data collected over a period of fourteen months. Exploratory data analysis (EDA) revealed distinct temporal patterns, including a "Social Jetlag" phenomenon characterized by reduced sleep on Mondays and compensatory rebound on Saturdays. The daily REM sleep duration was found to follow a Normal distribution, validated by the Kolmogorov-Smirnov (KS) test. Parameter estimation was conducted using Maximum Likelihood Estimation (MLE), Method of Moments (MoM), and Bayesian methods. A one-sample t-test confirmed that the subject's average daily REM sleep significantly differs from the physiological benchmark of 1.5 hours, specifically exceeding it. Finally, a linear regression model demonstrated a statistically significant, albeit weak, positive correlation between daily step count and REM sleep duration.**

***Index Terms*—statistical inference, exploratory data analysis, regression, hypothesis testing, sleep, rem sleep, physical activity**

I. Introduction

Sleep quality is a critical component of overall health, with Rapid Eye Movement (REM) sleep playing a vital role in memory consolidation and emotional processing. With the advent of wearable technology, it has become easier to track personal physiological metrics.

In healthy adults, REM sleep typically constitutes approximately 20–25% of total sleep duration [1]. As the data shows a mean sleep duration of 6.89 hours, this proportion corresponds to an expected REM sleep duration of approximately

$$6.89 \times 0.22 \approx 1.5 \text{ hours.}$$

This value is therefore used throughout this study as a physiological benchmark rather than a strict clinical recommendation.

The objective of this project is to apply statistical inference techniques to real-world data to answer two key questions:

1) Does the daily REM sleep duration follow a specific probability distribution?
2) Is there a significant relationship between physical exertion (measured by step count) and the duration of REM sleep obtained the subsequent night?

II. Dataset and Preprocessing

The dataset consists of 247 sleep observations collected between October 2024 and December 2025 using a Samsung Galaxy Watch.

A. Temporal Definitions

To avoid ambiguity, it is crucial to define the temporal assignment of sleep records. In this analysis, a "sleep record" is assigned to the day on which the subject wakes up.

- **Example:** Sleep occurring between Sunday 23.00 and Monday 07.00 is recorded as *Monday's* sleep duration.
- Consequently, behavioral factors described as "Sunday Night Procrastination" will manifest statistically in the *Monday* column.

B. Data Characteristics

The dataset spans over a year, capturing different seasonal and life-stage variations. Table I presents the monthly descriptive statistics for both REM sleep duration and physical activity.

TABLE I
Monthly Statistics: REM Sleep vs. Step Count

	REM Sleep Hours			Step Count		
Month-Year	N	Mean	Std	N	Mean	Std
10-2024	-	-	-	30	5,478	4,046
11-2024	29	1.90	0.59	30	7,732	3,940
12-2024	26	1.72	0.40	31	7,875	4,502
01-2025	15	1.66	0.56	31	6,780	4,181
02-2025	8	1.65	0.35	28	7,768	4,498
03-2025	4	1.98	0.30	31	6,954	4,377
04-2025	20	1.54	0.38	30	6,854	4,132
05-2025	19	1.47	0.44	30	6,645	3,781
06-2025	18	1.70	0.44	30	9,379	5,420
07-2025	21	1.75	0.24	31	8,618	4,393
08-2025	20	1.64	0.53	31	8,197	5,239
09-2025	18	1.77	0.48	30	9,944	5,126
10-2025	20	1.78	0.42	31	9,559	5,339
11-2025	20	1.56	0.60	29	9,763	5,102
12-2025	10	1.86	0.38	13	10,716	4,746

C. Preprocessing

To ensure a consistent unit of analysis, multiple sleep sessions (naps) occurring within the same day were aggregated to produce a single total sleep and REM duration for each date.

D. Variables

- **Dependent Variable (Y):** `daily_rem_hours` (Continuous). Calculated as the total minutes of REM sleep divided by 60.
- **Independent Variable (X):** `count` (Discrete). The total number of steps taken during the day.

E. Data Alignment

A critical preprocessing step involved aligning the timelines of the two datasets. Sleep data is timestamped by the "wake-up" time (end time), whereas step data is timestamped by the day the activity occurred. To analyze the effect of daily activity on that night's sleep, the step count from Day T was merged with the sleep record ending on Day $T + 1$.

After filtering for missing values and aligning dates, the final dataset contained $N = 247$ valid sample points.

III. Exploratory Data Analysis

To understand the underlying structure of the data, we computed summary statistics and visualized the distributions using histograms, boxplots, and scatter plots.

A. Summary Statistics

Table II summarizes the descriptive statistics of the key variables.

TABLE II
Summary Statistics

Statistic	Total Sleep Hours	REM Hours (Y)	Step Count (X)
N	247	247	247
Mean	6.89	1.70	8,094
Std. Dev	1.05	0.47	4,894
Min	4.17	0.42	18
Max	11.07	3.62	22,202
Skewness	0.94	0.26	-0.14
Kurtosis	2.20	0.73	-0.42

The skewness of 0.26 for REM sleep suggests a fairly symmetrical distribution, and the kurtosis of 0.73 indicates slightly heavier tails than a standard normal distribution but no extreme outliers.

B. Distribution Analysis

Fig. 1 displays the histogram of REM sleep hours with the MLE Normal fit overlaid. The visual evidence strongly supports a bell-shaped Normal distribution, centering around the 1.7 hour mark.

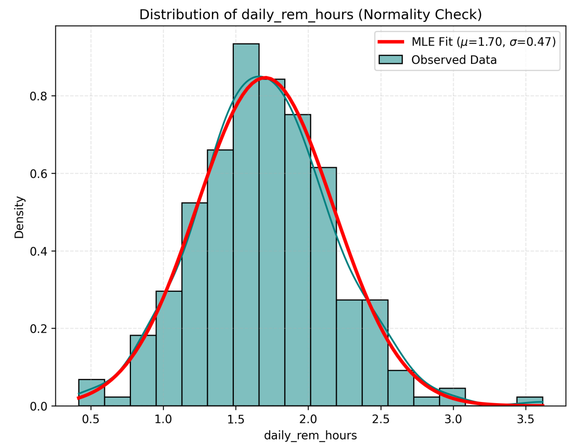

Fig. 1. Distribution of Daily REM Hours with MLE Normal Fit.

To further assess outliers and spread, we examined the boxplot in Fig. 2. The plot confirms the symmetry of the data, with the median centrally located within the interquartile range and minimal outliers.

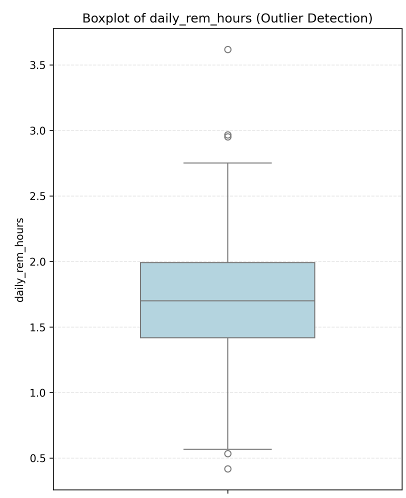

Fig. 2. Boxplot of Daily REM Sleep Hours.

C. Correlation Analysis

Fig. 3 illustrates the relationship between daily steps and REM sleep. A slight upward trend is visible, suggesting a positive correlation.

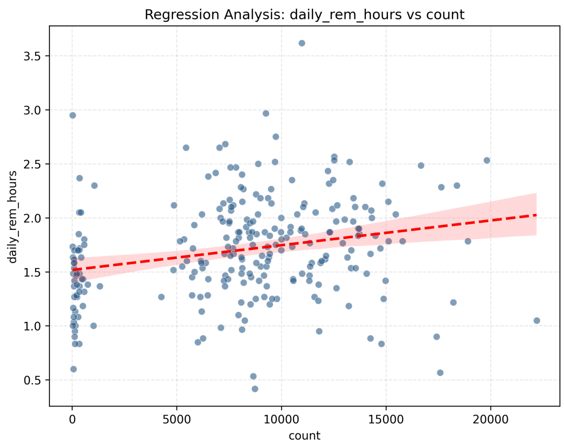

Fig. 3. Scatter Plot of REM Sleep vs. Daily Step Count with Regression Line.

D. Temporal and Behavioral Patterns

To investigate the impact of academic and work schedules on sleep architecture, we extended the exploratory analysis to include temporal trends.

1) Weekly Cycles (Social Jetlag): A distinct weekly cycle was observed, characteristic of "Social Jetlag." As shown in Table III, Mondays exhibit the lowest mean sleep (6.37 h) and REM duration (1.39 h).

This reduction can be attributed to behavioral "bedtime procrastination" on Sunday nights, a voluntary delay of sleep intended to psychologically extend the weekend and avoid the onset of the work week (often referred to as "Monday Syndrome"), resulting in curtailed rest despite the lack of a forced wake-up time. In contrast, Saturdays show a compensatory rebound effect, with the mean sleep duration peaking at 9.11 hours.

TABLE III
SLEEP STATISTICS BY WEEKDAY (HOURS)

Day	N	Total Sleep Mean	Total Sleep Std	REM Sleep Mean	REM Sleep Std
Monday	47	6.37	0.74	1.39	0.36
Tuesday	44	6.73	0.84	1.60	0.43
Wednesday	39	6.77	0.88	1.72	0.41
Thursday	48	6.79	0.82	1.76	0.36
Friday	45	6.87	0.81	1.76	0.45
Saturday	12	9.11	1.05	2.41	0.56
Sunday	13	7.96	1.49	2.05	0.52

Notably, Sundays exhibit the highest standard deviation in total sleep ($\sigma = 1.49$). This variance suggests that Sunday serves as a transitional period with irregular sleep behaviors, likely influenced by varying Monday schedules.

2) Physical Activity Patterns: To understand the drivers of the observed sleep variance, we analyzed daily step counts across the same period.

Table IV summarizes the weekly activity cycle. A clear behavioral pattern emerges: Weekdays (Monday–Friday) exhibit consistently higher mean step counts ($\mu > 8,700$), consistent with the commute to university classes (Tuesday). In contrast, Sundays show significantly reduced physical activity ($\mu \approx 2,458$), aligning with the "rest day" hypothesis and the potential measurement bias (reduced phone carriage).

TABLE IV
STEP COUNT STATISTICS BY WEEKDAY

Day	N	Mean Steps	Std Dev
Monday	62	8,754	2,924
Tuesday	62	9,739	4,278
Wednesday	63	9,264	3,831
Thursday	63	9,316	3,064
Friday	63	9,581	3,267
Saturday	62	7,059	5,830
Sunday	61	2,458	4,814

We also examined monthly trends in physical activity in Table I. The data indicates an increasing trend in activity towards the end of 2025, with December 2025 recording the highest mean step count ($\mu = 10,715$). Despite a reduced observation count relative to the recent months, this peak is attributed to a behavioral shift in the subject's commute, specifically the decision to walk directly to the tram station to avoid inconsistent bus wait times.

3) Long-Term Trends: Monthly analysis (Table I) reveals fluctuations in sleep duration. Notably, a dip in REM sleep duration is observed in May 2025 ($\mu = 1.47$).

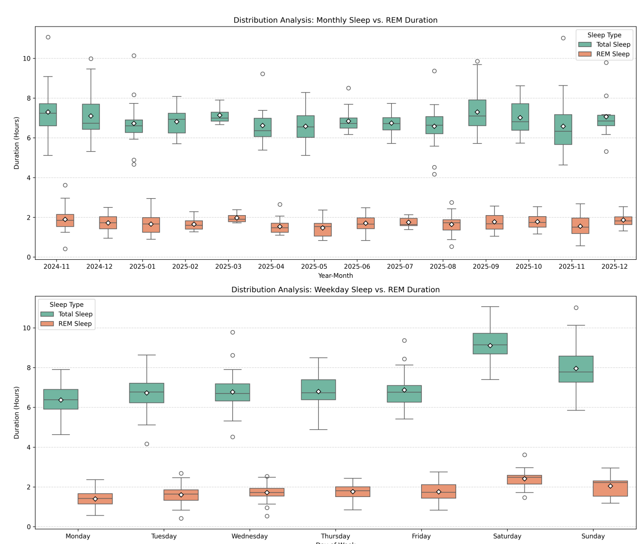

Fig. 4. Temporal Sleep Analysis: Showing monthly and weekday trends, respectively.

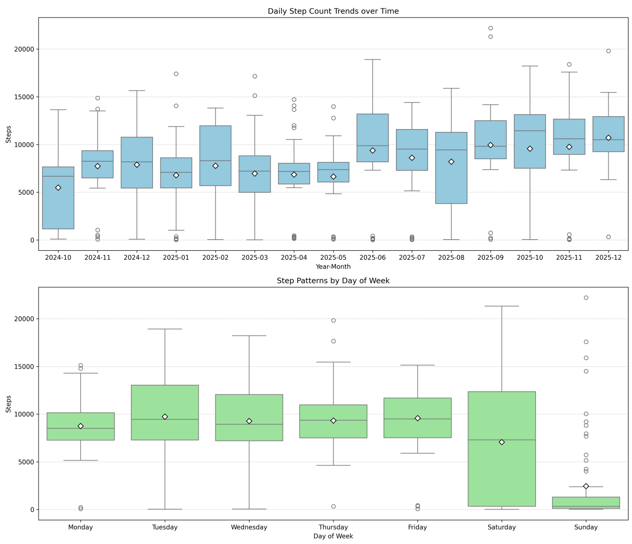

Fig. 5. Step Count Analysis: Showing monthly and weekday trends, respectively.

IV. Determining the Underlying Distribution

Based on the EDA, the Normal distribution was proposed as the underlying model for daily REM sleep.

A. Parameter Estimation

We estimated the parameters μ (mean) and σ (standard deviation) using three methods:

1) Maximum Likelihood Estimation: The MLE parameters were calculated as:

$$\hat{\mu}_{MLE} = 1.7023, \quad \hat{\sigma}_{MLE} = 0.4712$$

2) Method of Moments: For a Normal distribution, MoM estimates coincide with the sample moments:

$$\hat{\mu}_{MoM} = 1.7023, \quad \hat{\sigma}_{MoM} = 0.4712$$

3) Bayesian Estimation: We adopted a Bayesian framework with a conjugate Normal prior for the mean REM sleep duration. The prior mean was set to $\mu_0 = 1.5$ hours with prior standard deviation $\tau = 0.5$, representing a weakly informative prior centered at the benchmark value introduced earlier. The posterior mean is given by

$$\hat{\mu}_{Bayes} = \frac{\sigma^2 \mu_0 + n\tau^2 \bar{x}}{n\tau^2 + \sigma^2} = 1.7016. \quad (1)$$

The Bayesian estimate converges closely to the MLE due to the dominance of the likelihood ($N = 247$) over the prior.

V. Goodness-of-Fit and Hypothesis Testing

A. Goodness-of-Fit Test

To formally verify the assumption of Normality, we performed the Kolmogorov-Smirnov (KS) test.

- H_0**:** The data follows a Normal distribution.
- **Result:** $D = 0.0438$, p-value $= 0.7143$.

Since $p > 0.05$, we fail to reject the null hypothesis. This indicates that the data does not significantly deviate from a Normal distribution. Therefore, the assumption of normality is considered valid for the subsequent parametric analyses.

B. Hypothesis Testing on the Mean

We aimed to test if the subject's average REM sleep differs significantly from the benchmark of 1.5 hours.

$$H_0 : \mu = 1.5, \quad H_1 : \mu \neq 1.5$$

Since the data is Normal, we used a one-sample t-test.

- **T-Statistic:** 6.7334
- **P-Value:** < 0.0001

The result is statistically significant. We reject H_0 and conclude that the subject averages significantly more REM sleep than the 1.5 hour benchmark.

VI. Regression Analysis

To quantify the relationship between daily activity and REM sleep, we fitted a simple linear regression model:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where Y is `daily_rem_hours` and X is `count`.

A. Model Results

The regression output is summarized below:

- **Intercept (**β_0**):** 1.5169
  - t-statistic $= 26.799$
  - p-value < 0.001
- **Slope (**β_1**):** 2.29×10^{-5}
  - t-statistic $= 3.825$
  - p-value < 0.001
- **Model Fit:** $R^2 = 0.056$, $F = 14.63$ ($p < 0.001$)

B. Interpretation

The slope coefficient yields a t-statistic of 3.825, which corresponds to a p-value of effectively zero (< 0.001). This allows us to reject the null hypothesis ($\beta_1 = 0$) and conclude that there is a statistically significant positive relationship between step count and REM sleep.

However, the low R^2 value (0.056) indicates that daily steps explain only 5.6% of the variance in REM sleep. In practical terms, an increase of 10,000 steps correlates with approximately 14 minutes of additional REM sleep. While the relationship is significant, the effect size is modest, suggesting that sleep architecture is multifactorial and heavily influenced by unobserved variables.

VII. Conclusion

This study successfully applied statistical inference to personal health data. We confirmed that daily REM sleep duration follows a Normal distribution. Hypothesis testing showed the subject averages significantly more REM sleep than the standard benchmark. Finally, regression analysis established a positive, statistically significant link between daily step count and REM sleep, confirming the initial hypothesis that daily activity impacts sleep quality, though the effect size is modest.

ACKNOWLEDGEMENT

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REFERENCES

[1] M. A. Carskadon and W. C. Dement, "Normal human sleep: An overview," in *Principles and Practice of Sleep Medicine*, 6th ed., M. H. Kryger, T. Roth, and W. C. Dement, Eds. Philadelphia, PA, USA: Elsevier, 2017, pp. 15–24.